



3 Complex Analysis 2024

Tutorial 1

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Complex numbers: basics

Problem 1 Find the real and the imaginary parts of the complex numbers

$$z_1 = (3 + 2i)^3 \quad \text{and} \quad z_2 = \frac{i + 1}{i - 1}.$$

Solution. Using the binomial formula, we obtain

$$\begin{aligned} z_1 &= (3 + 2i)^3 = 3^3 + 3 \cdot 3^2 \cdot (2i) + 3 \cdot 3 \cdot (2i)^2 + (2i)^3 \\ &= 27 + 54i - 36 - 8i = -9 + 46i, \end{aligned}$$

i.e. $\operatorname{Re} z_1 = -9$, $\operatorname{Im} z_1 = 46$. For z_2 we have

$$z_2 = \frac{i + 1}{i - 1} = \frac{(i + 1)(i + 1)}{(i - 1)(i + 1)} = \frac{i^2 + 2i + 1}{i^2 - 1^2} = \frac{-1 + 2i + 1}{-1 - 1} = \frac{2i}{-2} = -i,$$

so that $\operatorname{Re} z_2 = 0$ and $\operatorname{Im} z_2 = -1$.

Problem 2 Prove the following inequalities for $z, w \in \mathbb{C}$:

- (a) $\max\{|\operatorname{Re} z|, |\operatorname{Im} z|\} \leq |z| \leq |\operatorname{Re} z| + |\operatorname{Im} z|,$
- (b) $||w| - |z|| \leq |w - z|.$

Solution.

- (a) Let $z = x + iy$ with $\operatorname{Re} z = x$, $\operatorname{Im} z = y$. We have

$$|z| = \sqrt{x^2 + y^2} \geq \sqrt{x^2} = |x| \quad \text{and} \quad |z| = \sqrt{x^2 + y^2} \geq \sqrt{y^2} = |y|.$$

This gives $\max\{|\operatorname{Re} z|, |\operatorname{Im} z|\} = \max\{|x|, |y|\} \leq |z|$. For the second inequality, we consider equivalent inequalities

$$\begin{aligned} |z| \leq |\operatorname{Re} z| + |\operatorname{Im} z| &\iff \sqrt{x^2 + y^2} \leq |x| + |y| \iff x^2 + y^2 \leq (|x| + |y|)^2 \\ &\iff x^2 + y^2 \leq x^2 + 2|xy| + y^2 \iff 0 \leq 2|xy|. \end{aligned}$$

The last inequality is trivially true, this implies that also the first inequality is true.

(b) Using the triangle inequality, we estimate

$$|w| = |w - z + z| \leq |w - z| + |z|,$$

and this gives

$$|w| - |z| \leq |w - z|.$$

Interchanging z and w , we obtain

$$|z| - |w| \leq |w - z|.$$

The last two inequalities together give

$$||w| - |z|| \leq |w - z|.$$

Problem 3 Find all $z \in \mathbb{C}$ with the following property:

(a) $z = |z|$,

(b) $|z - i| = 2z + i$.

Solution.

(a) Let $z = x + iy$ with $\operatorname{Re} z = x$, $\operatorname{Im} z = y$. The equation $z = |z|$ is equivalent to

$$x + iy = \sqrt{x^2 + y^2}.$$

Equating the real and the imaginary parts, we obtain

$$x = \sqrt{x^2 + y^2}, \quad y = 0.$$

In particular, z is real. Substituting $y = 0$ in the first equation, we see that

$$x = \sqrt{x^2} = |x|,$$

and this implies that $x \geq 0$. Thus, the set of $z \in \mathbb{C}$ with $z = |z|$ is exactly the set of non-negative real numbers.

(b) We know that the modulus of any number is a non-negative real number. Therefore, let

$$|z - i| = 2z + i = a,$$

where $a \in \mathbb{R}$ and $a \geq 0$. Then

$$z = \frac{1}{2}a - \frac{1}{2}i,$$

and

$$|z - i| = \left| \frac{1}{2}a - \frac{3}{2}i \right| = \sqrt{\left(\frac{1}{2}a\right)^2 + \left(-\frac{3}{2}\right)^2} = \sqrt{\frac{a^2 + 9}{4}} = \frac{\sqrt{a^2 + 9}}{2} = a,$$

so that

$$\sqrt{a^2 + 9} = 2a.$$

This is equivalent to $a^2 + 9 = 4a^2$, i.e. $a^2 = 3$. Taking into account that $a \geq 0$, we arrive at $a = \sqrt{3}$. Thus, the only solution of the equation $|z - i| = 2z + i$ is

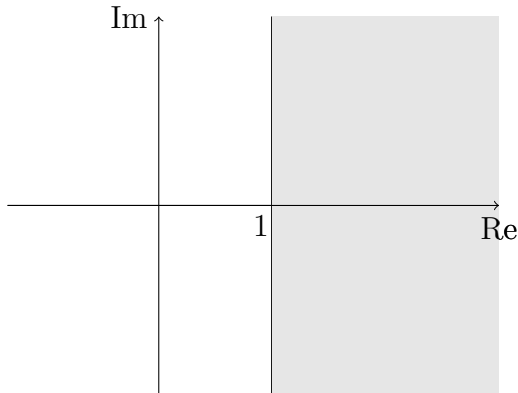
$$z = \frac{\sqrt{3}}{2} - \frac{1}{2}i.$$

Problem 4 Sketch the following sets in the complex plane:

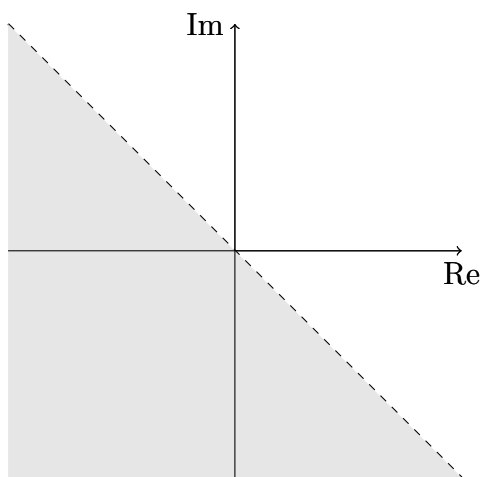
- (a) $\{z \in \mathbb{C} : \operatorname{Re}(z - 1) \geq 0\}$,
- (b) $\{z \in \mathbb{C} : \operatorname{Im}((1 + i)z) < 0\}$,
- (c) $\{z \in \mathbb{C} : 2 < |z| < 4\}$,
- (d) $\{z \in \mathbb{C} : |z - i| \leq 2 \text{ and } \operatorname{Im} z \geq 2\}$.

Solution.

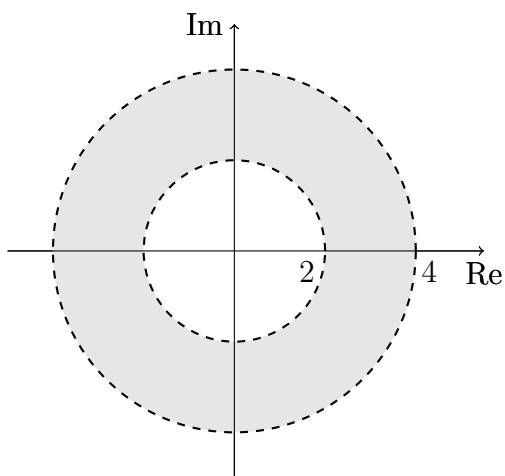
- (a) Let $z = x + iy$, then $\operatorname{Re}(z - 1) \geq 0$ is equivalent to $x - 1 \geq 0$.



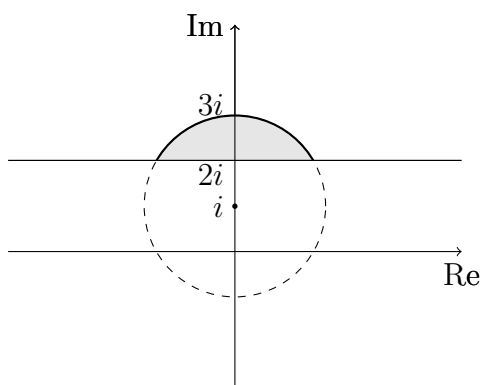
- (b) Let $z = x + iy$, then $(1 + i)z = (1 + i)(x + iy) = (x - y) + i(x + y)$, so that $\operatorname{Im}((1 + i)z) = x + y$. The set corresponds to the set of points (x, y) in the Cartesian plane with $x + y < 0$.



(c)



(d)



Problem 5 Consider the equation

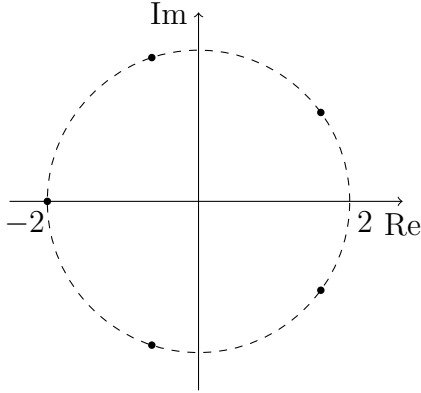
$$z^5 + 32 = 0.$$

(a) Plot the solutions of this equation in the complex plane.

- (b) Give the solutions in polar coordinates.

Solution.

- (a) The solutions are the 5th roots of -32 . The real solution is -2 , and the complex solutions are uniformly distributed on the circle with radius 2.



- (b) We have $z^5 = -32 = 2^5 \cdot (\cos(\pi + 2\pi n) + i \sin(\pi + 2\pi n))$ with $n \in \mathbb{Z}$. By de Moivre's Theorem, the roots have the form

$$z = 2 \cdot \left(\cos \frac{\pi + 2\pi n}{5} + i \sin \frac{\pi + 2\pi n}{5} \right), \quad n \in \mathbb{Z}.$$

Taking $n = 0, 1, 2, 3, 4$, we obtain for the roots

$$\begin{aligned} z_0 &= 2 \cdot \left(\cos \frac{\pi}{5} + i \sin \frac{\pi}{5} \right), \\ z_1 &= 2 \cdot \left(\cos \frac{3\pi}{5} + i \sin \frac{3\pi}{5} \right), \\ z_2 &= 2 \cdot \left(\cos \frac{5\pi}{5} + i \sin \frac{5\pi}{5} \right) = -2, \\ z_3 &= 2 \cdot \left(\cos \frac{7\pi}{5} + i \sin \frac{7\pi}{5} \right), \\ z_4 &= 2 \cdot \left(\cos \frac{9\pi}{5} + i \sin \frac{9\pi}{5} \right). \end{aligned}$$